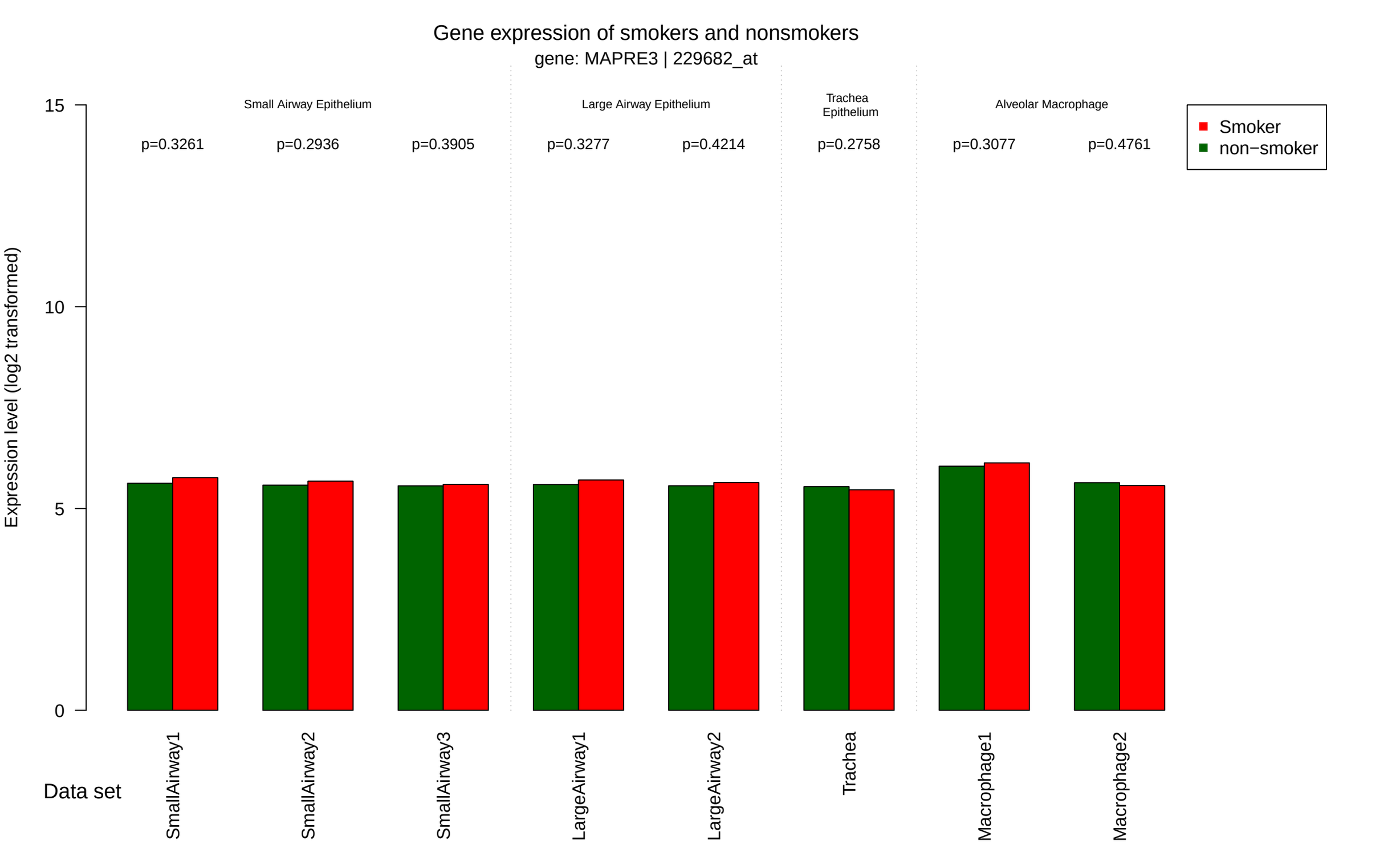
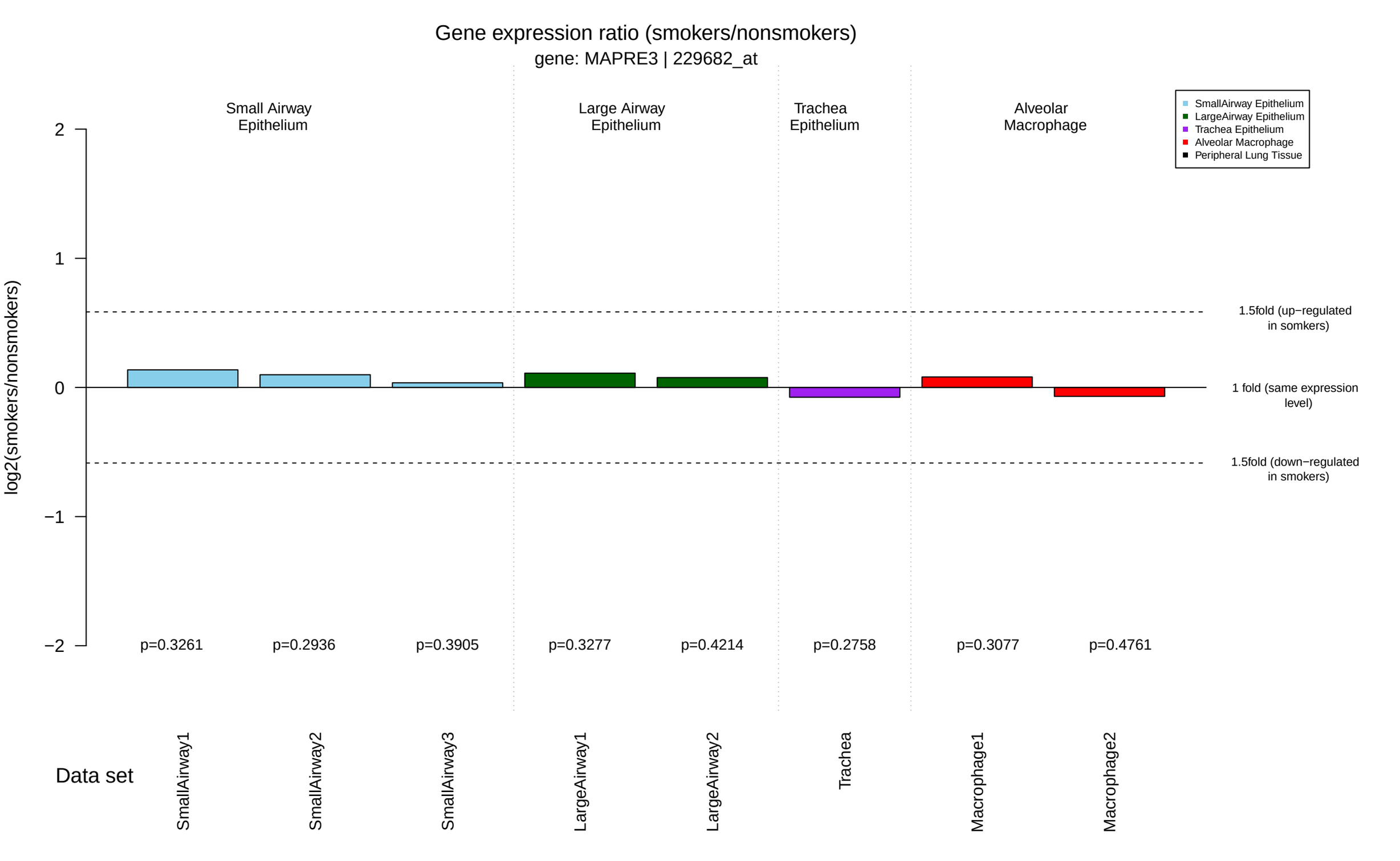
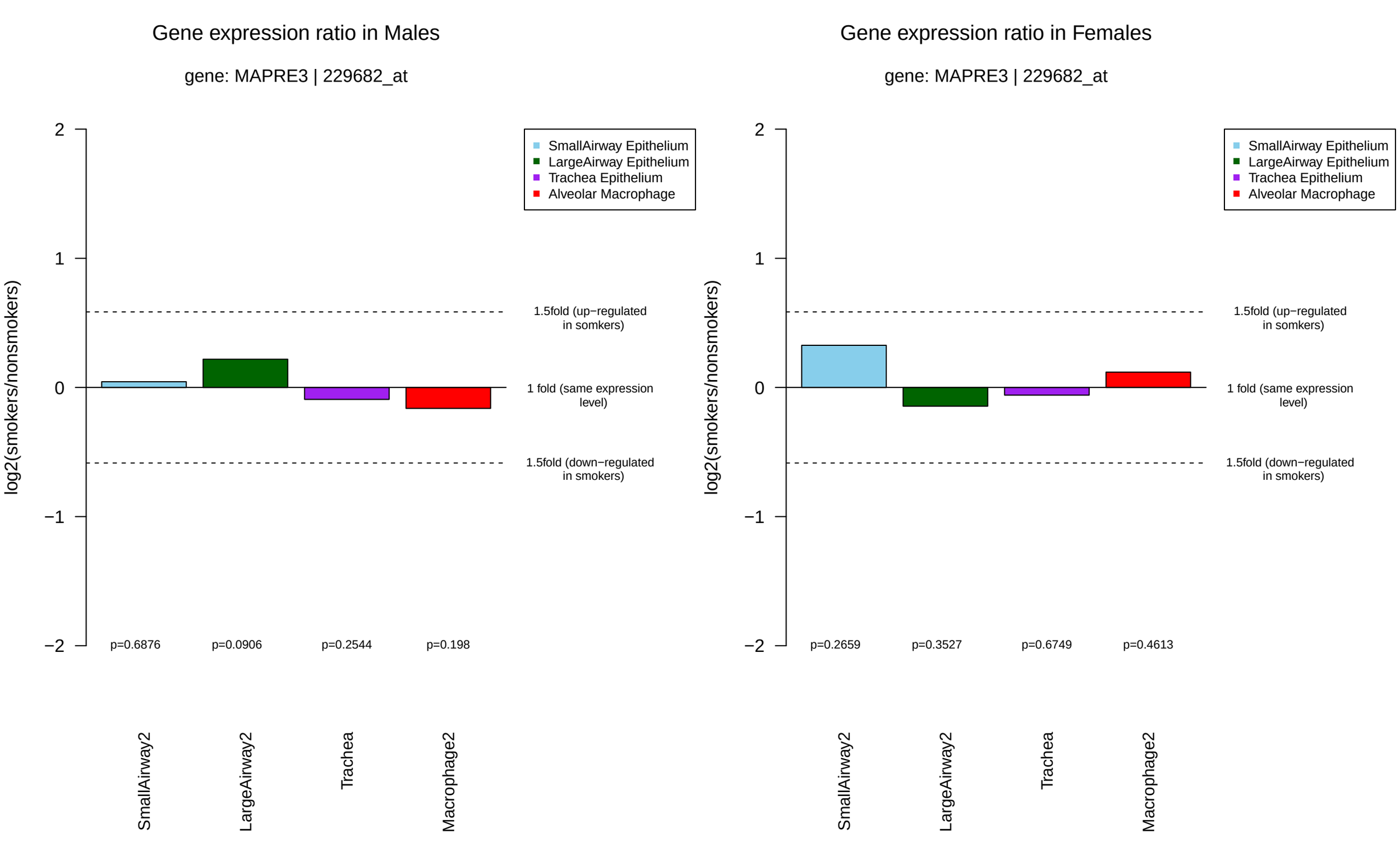
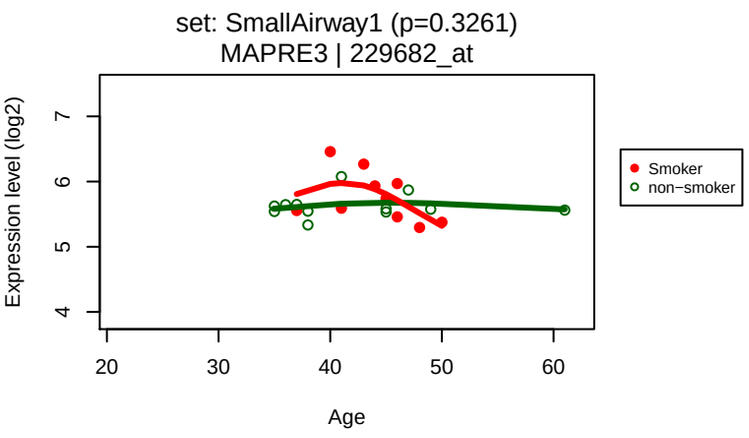
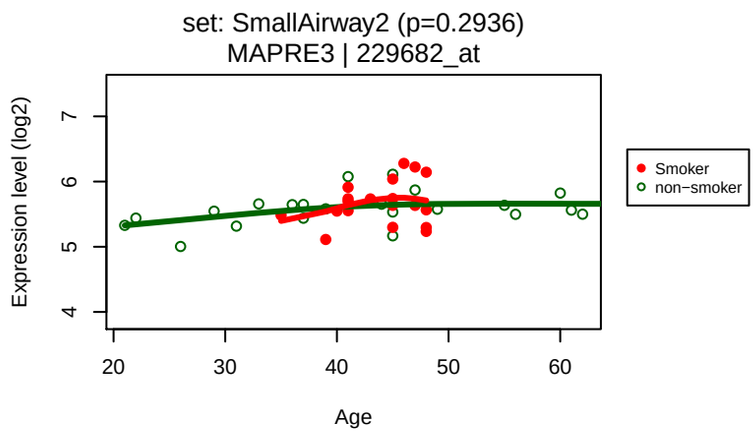
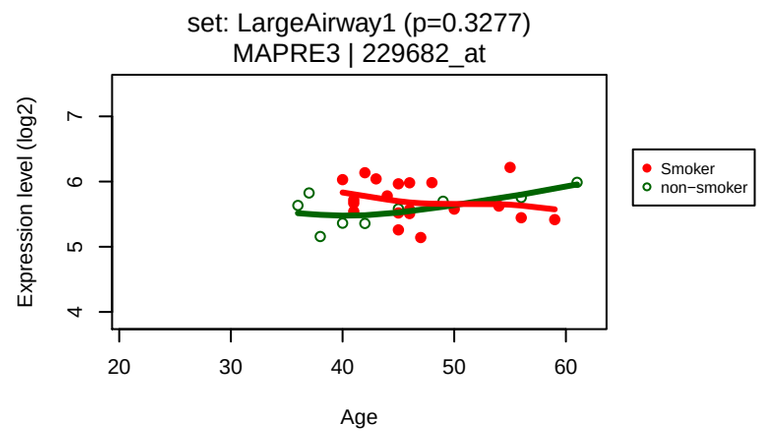

Gene expression of smokers and nonsmokers
gene: MAPRE3 | 229682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3261
p=0.2936
p=0.3905
p=0.3277
p=0.4214
p=0.2758
p=0.3077
p=0.4761
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

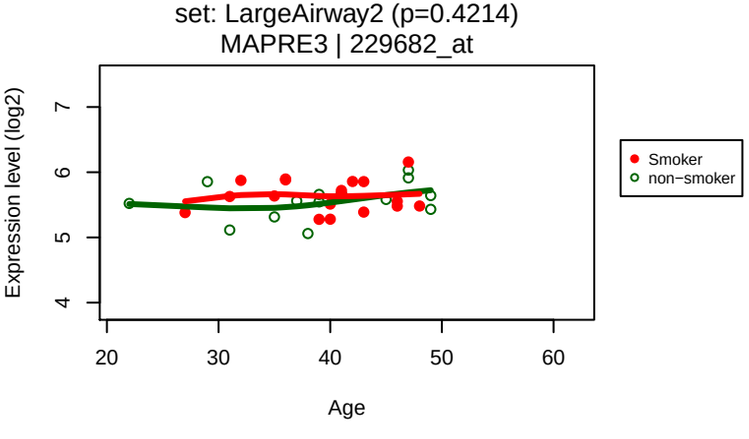

Gene expression ratio (smokers/nonsmokers)
gene: MAPRE3 | 229682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3261
p=0.2936
p=0.3905
p=0.3277
p=0.4214
p=0.2758
p=0.3077
p=0.4761
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

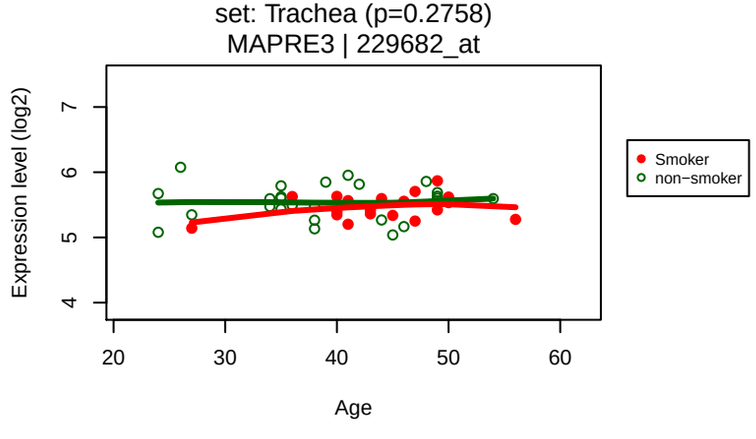

Gene expression ratio in Males
gene: MAPRE3 | 229682_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6876
p=0.0906
p=0.2544
p=0.198
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPRE3 | 229682_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2659
p=0.3527
p=0.6749
p=0.4613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

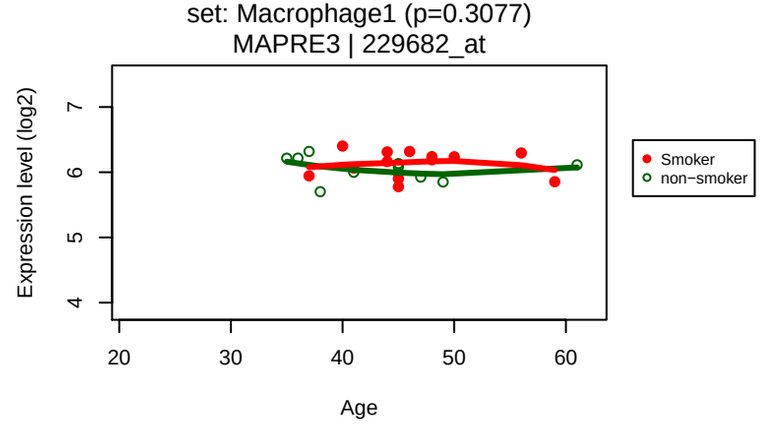

set: SmallAirway1 (p=0.3261)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker

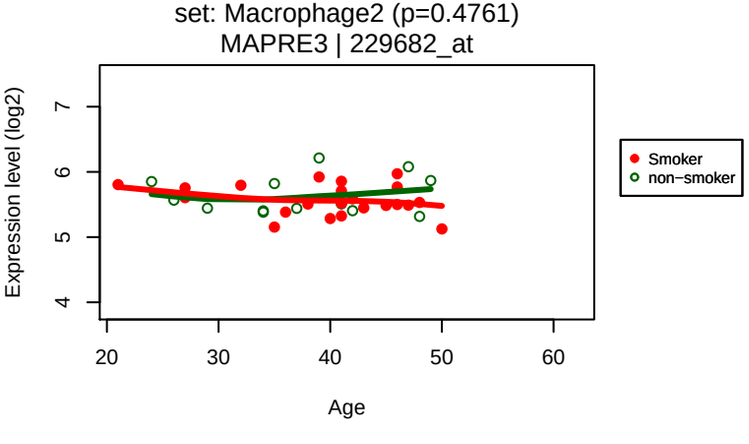

set: SmallAirway2 (p=0.2936)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker

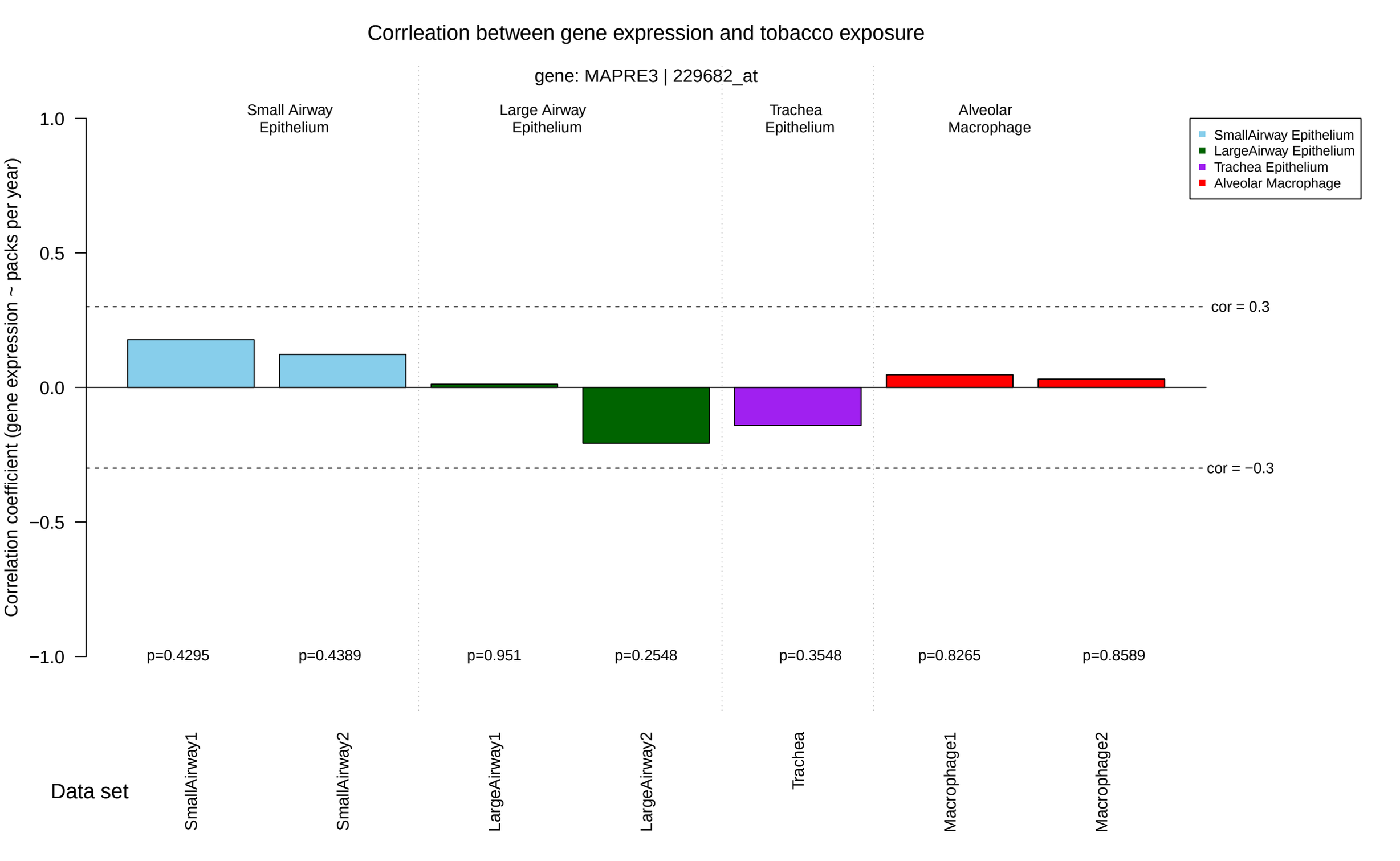

set: LargeAirway1 (p=0.3277)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4214)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2758)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3077)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4761)
MAPRE3 | 229682_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPRE3 | 229682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4295
p=0.4389
p=0.951
p=0.2548
p=0.3548
p=0.8265
p=0.8589
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2